HDOC DAY – 74

1. Create an enumeration for months and print how many days each month has.
    a. Input: FEB
    b. Output: 28 or 29 days
2. Define an enumeration with SUCCESS, FAILURE, and TIMEOUT, and print messages accordingly.
    a. Input: FAILURE
    b. Output: Operation Failed

1. Program to Create an Enumeration for Months and Print the Number of Days

Algorithm

1. Start the program.
2. Include the required header files stdio.h and string.h.
3. Declare an enumeration containing all 12 months.
4. Declare a character array to accept the month name.
5. Read the month from the user.
6. If the month is FEB, display "28 or 29 days".
7. If the month is APR, JUN, SEP or NOV, display "30 days".
8. Otherwise, display "31 days".
9. Stop the program.

C Program

```
#include <stdio.h>
#include <string.h>

enum Month {
   JAN, FEB, MAR, APR, MAY, JUN,
   JUL, AUG, SEP, OCT, NOV, DEC
};
```

```
int main() {
    char month[10];

    printf("Enter month: ");
    scanf("%s", month);

    if (strcmp(month, "FEB") == 0)
        printf("28 or 29 days");
    else if (strcmp(month, "APR") == 0 ||
             strcmp(month, "JUN") == 0 ||
             strcmp(month, "SEP") == 0 ||
             strcmp(month, "NOV") == 0)
        printf("30 days");
    else
        printf("31 days");

    return 0;
}
```

Test Case Table

Test Case	Input	Expected Output
1	JAN	31 days
2	FEB	28 or 29 days
3	APR	30 days
4	JUN	30 days
5	AUG	31 days
6	NOV	30 days

7	DEC	31 days

Execution / Output

Execution 1:

```
Enter month: FEB
[28 or 29 days%
```

Execution 2:

```
Enter month: APR
30 days%
```

2. Program to Define an Enumeration with SUCCESS, FAILURE and TIMEOUT

Algorithm

1. Start the program.
2. Include the required header files stdio.h and string.h.
3. Define an enumeration named Status.
4. Declare three constants: SUCCESS, FAILURE and TIMEOUT.
5. Declare a character array to accept the status.
6. Read the status from the user.
7. Compare the input with SUCCESS, FAILURE and TIMEOUT.
8. If the input is SUCCESS, display "Operation Successful".
9. If the input is FAILURE, display "Operation Failed".
10. If the input is TIMEOUT, display "Operation Timed Out".
11. If the input does not match any status, display "Invalid Status".
12. Stop the program.

C Program

```
#include <stdio.h>
#include <string.h>

enum Status {
```

```
    SUCCESS,
    FAILURE,
    TIMEOUT
};

int main() {
    char input[20];
    enum Status status;

    printf("Enter status: ");
    scanf("%s", input);

    if (strcmp(input, "SUCCESS") == 0) {
        status = SUCCESS;
        printf("Operation Successful");
    }
    else if (strcmp(input, "FAILURE") == 0) {
        status = FAILURE;
        printf("Operation Failed");
    }
    else if (strcmp(input, "TIMEOUT") == 0) {
        status = TIMEOUT;
        printf("Operation Timed Out");
    }
    else {
        printf("Invalid Status");
    }

    return 0;
}
```

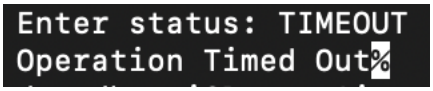
Test Case Table

Test Case	Input	Expected Output
1	SUCCESS	Operation Successful
2	FAILURE	Operation Failed

3	TIMEOUT	Operation Timed Out
4	ERROR	Invalid Status

Execution / Output

Execution 1:



Execution 2:



Result

Both programs were successfully implemented and executed using enumeration concepts in C. The first program determines the number of days in a given month, while the second program displays appropriate messages based on the operation status (SUCCESS, FAILURE or TIMEOUT).